

Generation Interconnection System Impact Study Report

For Queue Project Q24-2429

Truman 230 kV
December 03, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2429. The Interconnection Customer proposes a 10.0 MW solar photovoltaic facility interconnecting at the Truman 230 kV substation in the Ohio area. The request was received on July 09, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 393 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$184,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PORTAGE LAKES – WEEDPATCH 230 KV	HYLEBOS – RAVENNA 500 KV line	111.3	1,195	5.0
CANNON BRANCH – GALION GENERATING STATION 230 KV	TREGO – BEAVER 500 KV line	107.8	643	6.0
CANNON BRANCH – MCMILLAN 230 KV	TREGO SUBSTATION – BEAVER SUBSTATION 500 KV	104.4	653	6.0
W MILLERSBURG – CANAL RD 230 KV	CANAL ROAD – WEEDPATCH 230 KV line	102.0	574	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEEDPATCH – PORTAGE LAKES 230 KV	Rebuild line with higher-capacity conductor	\$26,081,000	0% (\$77,000)
CANNON BRANCH – GALION GENERATING STATION 230 KV	Reconductor line	\$7,195,000	1% (\$42,000)
CANNON BRANCH – MCMILLAN SUBSTATION 230 KV	Reconductor line	\$5,373,000	1% (\$31,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W MILLERSBURG – CANAL RD 230 KV	Reconductor line	\$14,954,000	0% (\$34,000)
	Total allocated to project		\$184,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.